



Introduction to simple linear equations. Systems of Linear equations.

Lesson 1

Systems of Equations

- ▶ A set of equations is called a **system of equations**.
- ▶ The **solutions** must satisfy each equation in the system.
- ▶ If all equations in a system are linear, the system is a **system of linear equations**, or a **linear system**.



The form of a linear equation depends on the number of dimensions of the space it belongs to.

- For instance, in a 2 dimensional space (i.e. a rectangular xy – coordinate system) the equation of a line is usually written as:
 $ax + by = c$ (a, b not both 0) (1)
- In a 3 dimensional space (i.e. in a rectangular xyz – coordinate system), the equation of a plane is usually written as:
 $ax + by + cz = d$ (a, b, c not all 0) (2)



In n- dimensional space...

- Generalizing from the above two examples, we can conclude that the general form of a equation in a “n” dimensional space can be written as:
- $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$,
where $a_1, \dots, a_n$ are coefficients and not all 0 (3)



Special case

- When $b = 0$, equation (3) is called a homogeneous linear equation, and can be written as follows:
- $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$ (4)

Examples of linear equations

- ▶ The following equations are linear equations. We recognize and identify them as linear because they:
- ▶ Do not contain any products or roots of the variables;
- ▶ Variables do not contain any powers bigger than 1 ;
- ▶ Variables do not appear as the arguments of a trigonometric, logarithmic or exponential function.
- ▶ $2x - 3y = 5$
- ▶ $\frac{1}{3}x - y = \frac{5}{7}$
- ▶ $x_1 + 2x_2 - 3x_3 = 0$



Are these linear equations?

➤ $2x - 3y^2 = 7$

➤ $\sin x + 3y = 0$

➤ $3x - 2\sqrt{y} = 1$

➤ $\log x = 3y$

General form of a system of m equations with n variables

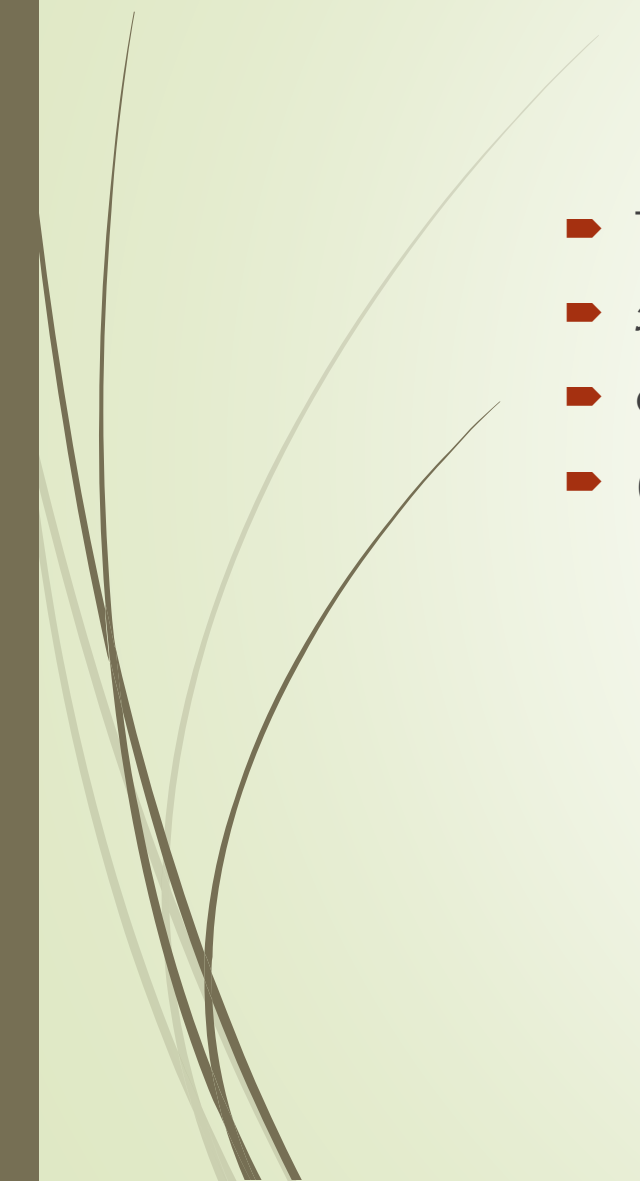
- A set of linear equations is called a system of linear equations or a linear system.
- A general system of m linear equations with n unknowns (or variables) can be written as:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

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- The solutions can be written as:
 - $x_1 = s_1$, $x_2 = s_2$...etc.
 - or
 - $(s_1, s_2, \dots, s_n)$



Types of Solutions for systems of linear equations with 2 variables

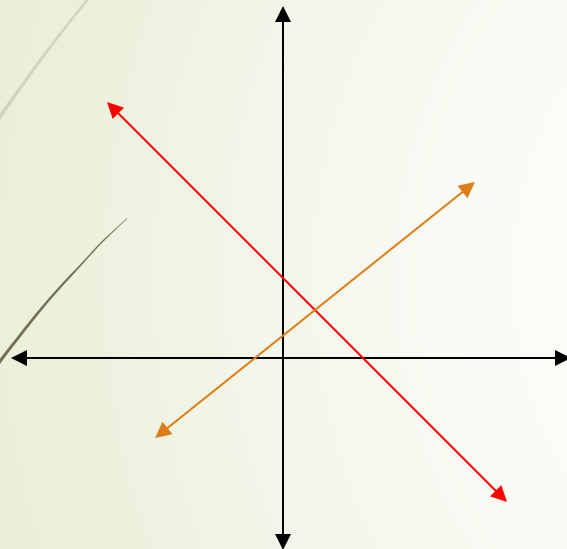
A solution to a system of equations is an **ordered pair** that satisfy all the equations in the system.

A system of linear equations can have:

- ➡ 1. Exactly one solution
- ➡ 2. No solutions
- ➡ 3. Infinitely many solutions

Graphic representation of solutions for Linear Systems of 2 equations

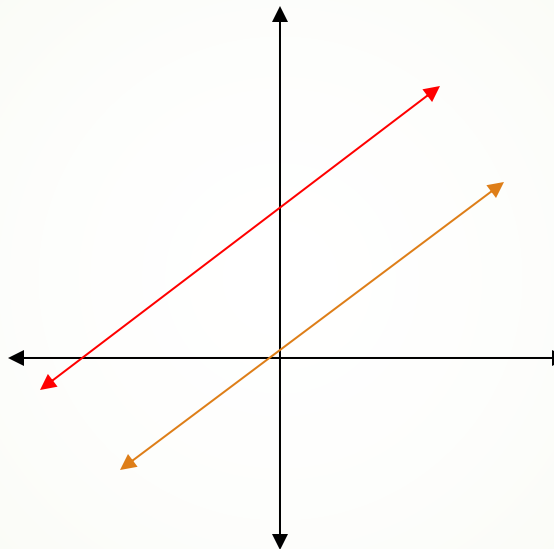
Consistent



One solution

Lines intersect

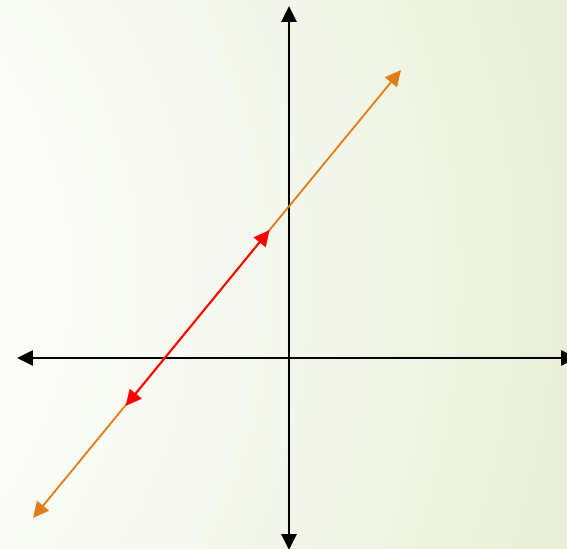
Inconsistent



No solution

Lines are parallel

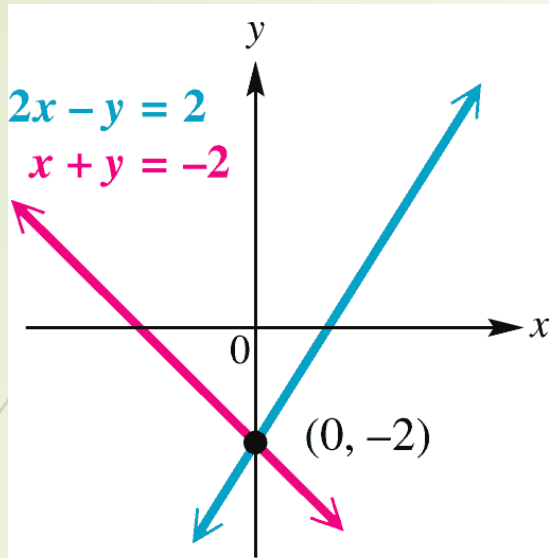
Dependent



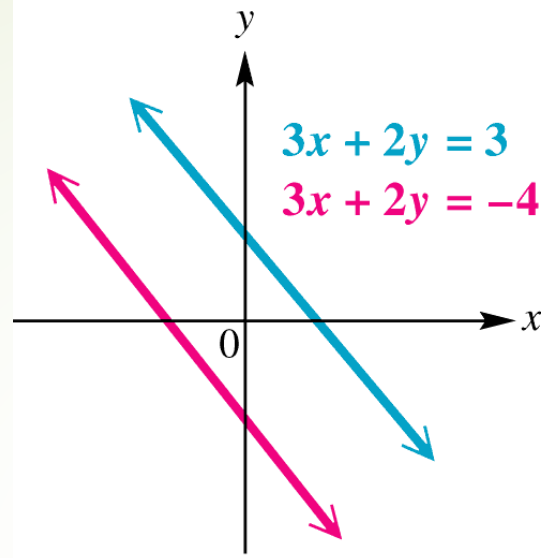
Infinite number of
solutions

Coincide-Same
line

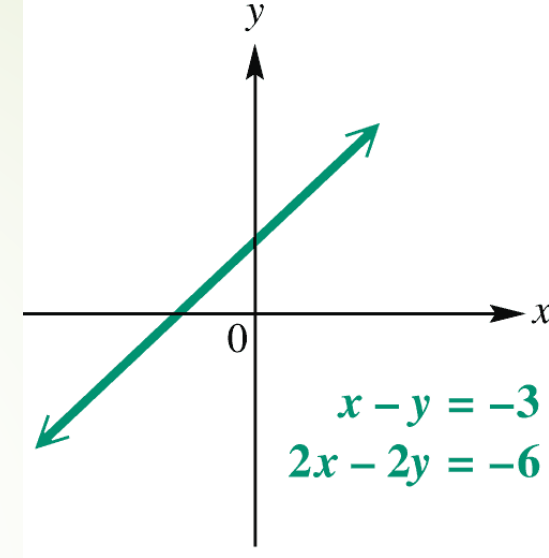
Linear System in Two Variables -



One Solution



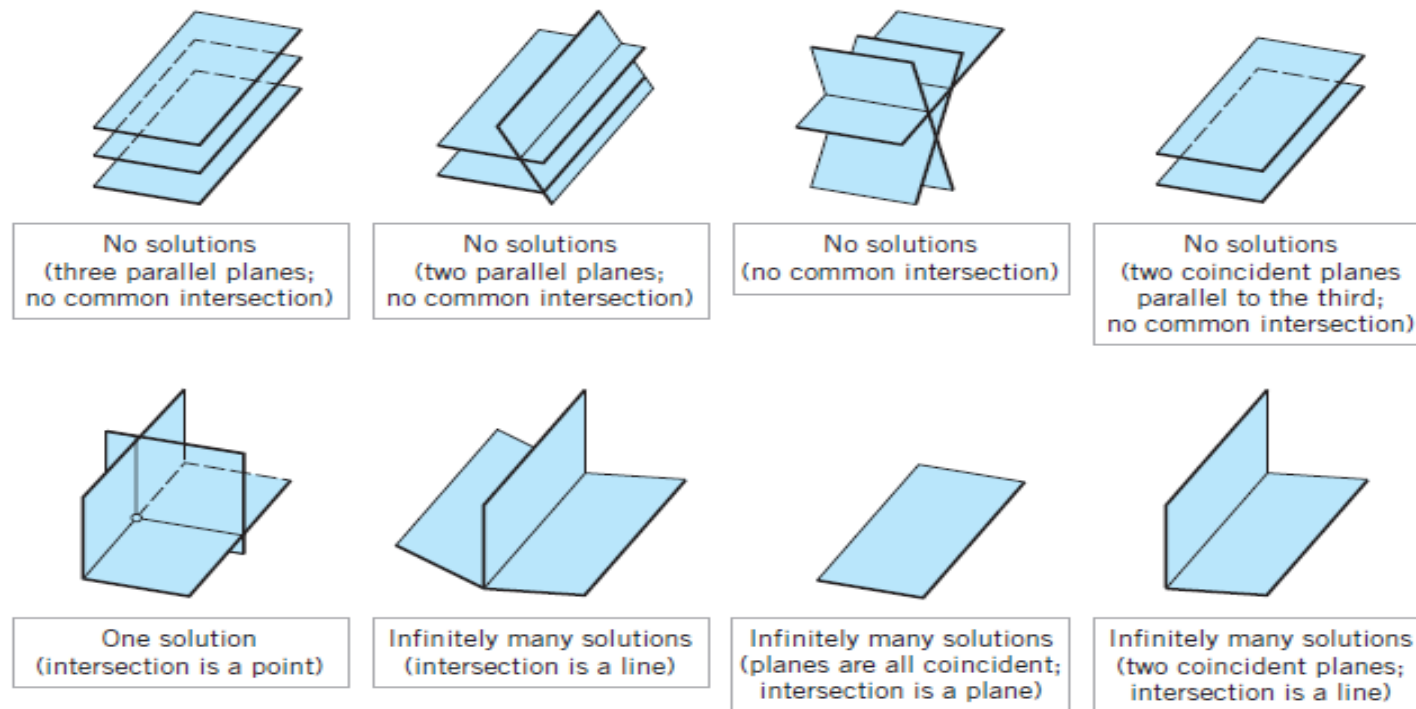
No Solutions



Infinitely Many Solutions

- Three possible solutions to a linear system in two variables:
 - One solution: coordinates of a point
 - No solutions: **inconsistent** case
 - Infinitely many solutions: **dependent** case

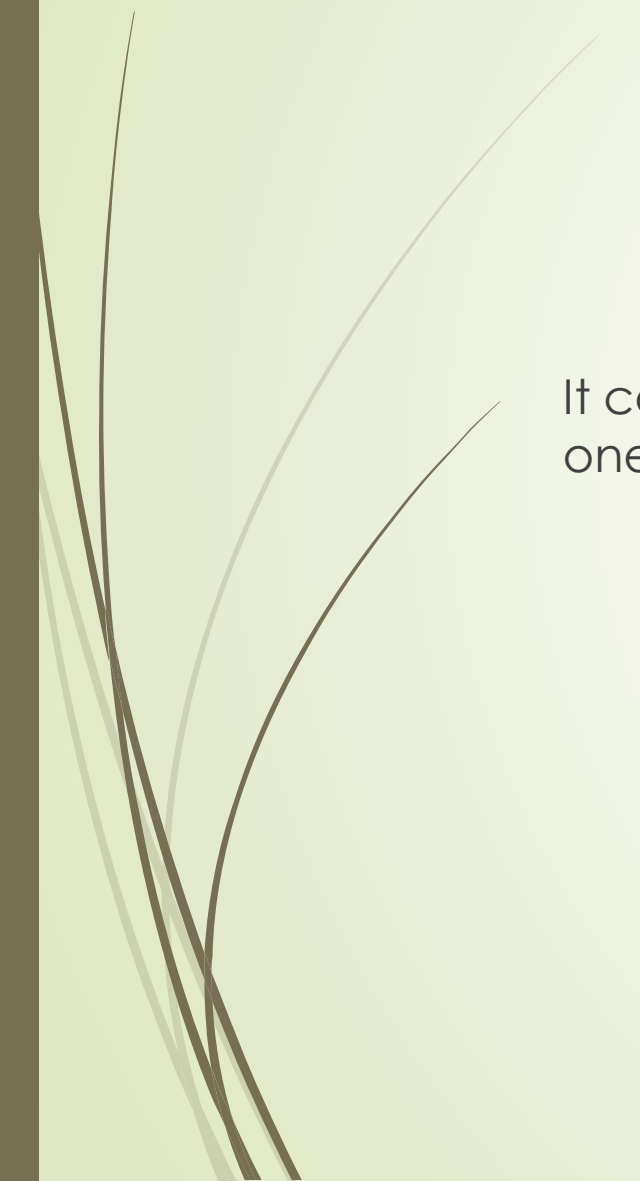
Graphic representation for types of solutions of Linear Systems in Three Variables



▲ Figure 1.1.2



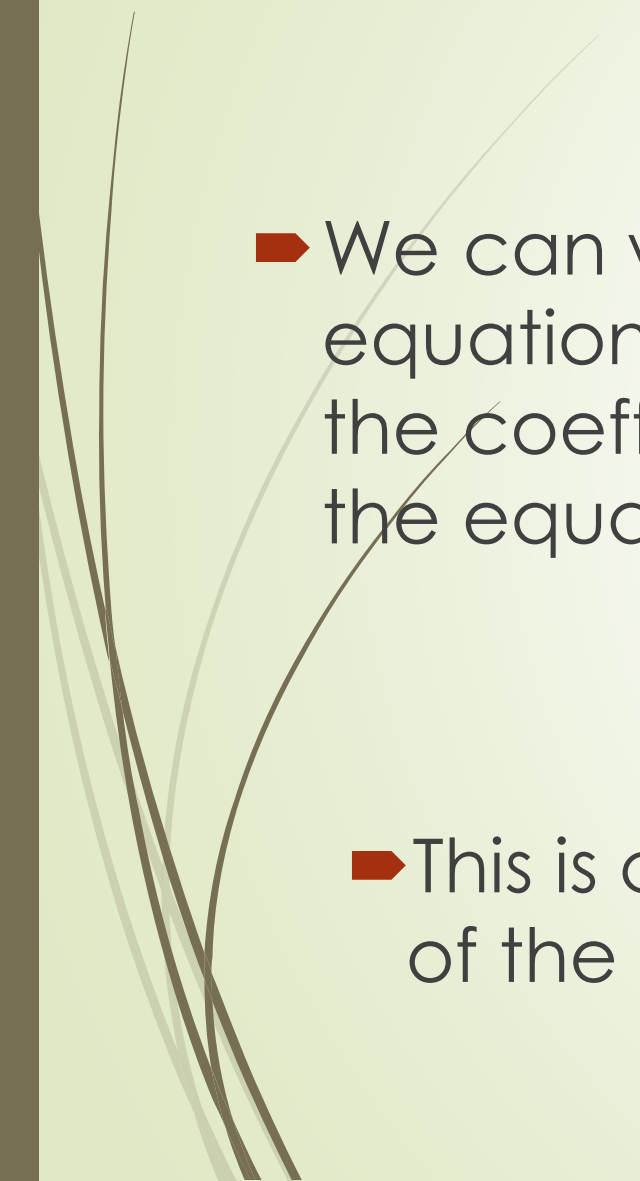
For systems with n variables...



It can be proved that any linear system (of any number of variables) has zero, one, or infinitely many solutions. There is no other possibility.



Augmented Matrix

- We can write a system of linear equations as a matrix by writing only the coefficients and constants that appear in the equations.
 - This is called the augmented matrix of the system.
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Augmented Matrix

Linear System	Augmented Matrix
$\begin{cases} 3x - 2y + z = 5 \\ x + 3y - z = 0 \\ -x + \quad 4z = 11 \end{cases}$	$\begin{bmatrix} 3 & -2 & 1 & 5 \\ 1 & 3 & -1 & 0 \\ -1 & 0 & 4 & 11 \end{bmatrix}$

- Notice that a missing variable in an equation corresponds to a 0 entry in the augmented matrix.

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- Write the augmented matrix of the system of equations.

Example—Finding Augmented Matrix of Linear System

$$\begin{cases} 6x - 2y - z = 4 \\ x + 3z = 1 \\ 7y + z = 5 \end{cases}$$



Elementary Row Operations

1. Interchange two rows.
 2. Multiply/divide a row by a nonzero constant.
 3. Add/subtract 2 rows.
- ➡ Note that performing any of these operations on the augmented matrix of a system does not change its solution.

Example: Solving a system of 2 linear equations using the augmented matrix and row operations

Linear system		Augmented matrix
$2x - y = 5$	Equation 1	$\begin{bmatrix} 2 & -1 & 5 \\ 1 & 4 & 7 \end{bmatrix}$
$x + 4y = 7$	Equation 2	
		$\begin{matrix} x & y \end{matrix}$

- This matrix is called the augmented matrix of the system.
- The augmented matrix contains the same information as the system, but in a simpler form.
- The operations we learned for solving systems of equations can now be performed on the augmented matrix.



Practice row operations to solve the following system

➤ $x + y + 2z = 9$

➤ $2x + 4y - 3z = 1$

➤ $3x + 6y - 5z = 0$

1	1	2	9
2	4	-3	1
3	6	-5	0

